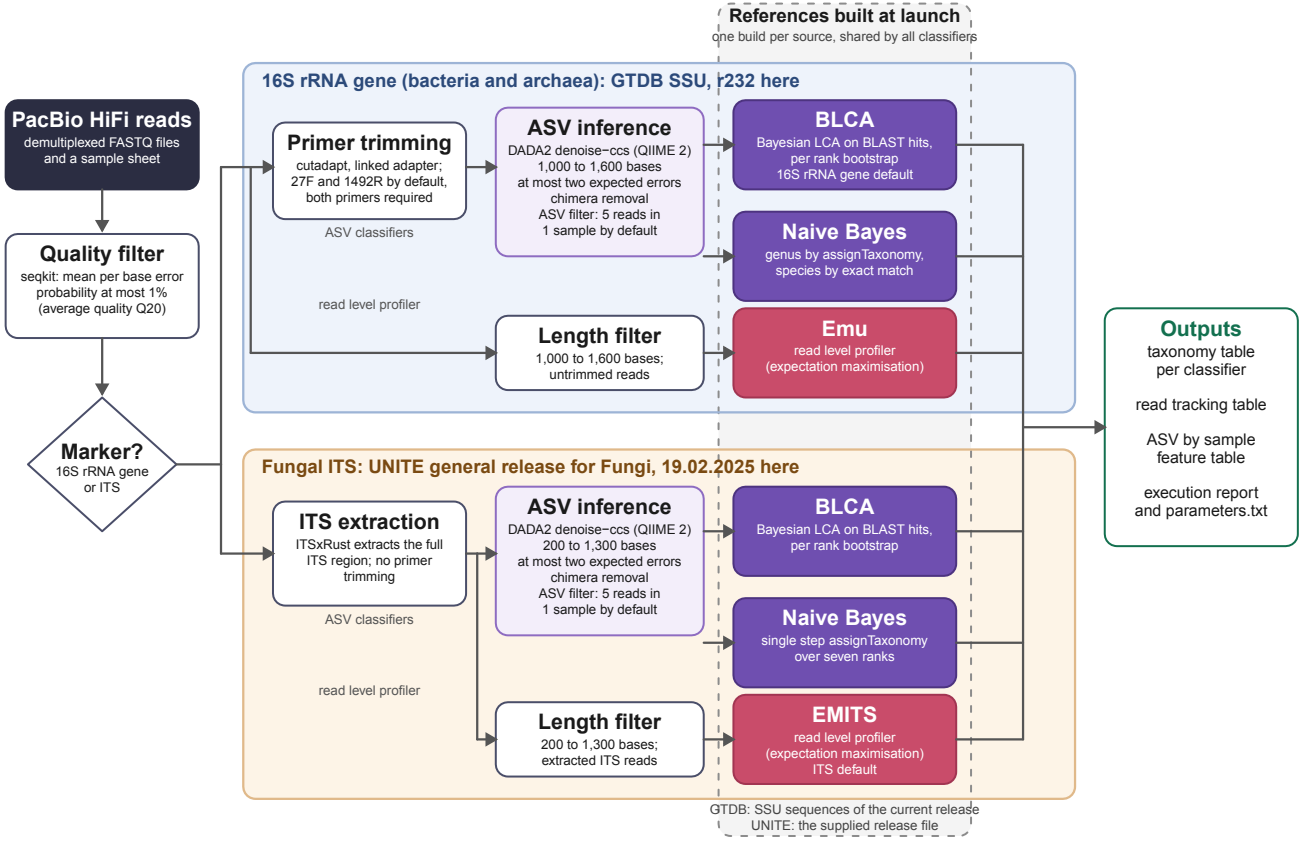
References built at launch
one build per source, shared by all classifiers
PacBio HiFi reads
demultiplexed FASTQ files and a sample sheet
Quality filter
seqkit: mean per base error probability at most 1% (average quality Q20)
Marker?
16S rRNA gene or ITS
16S rRNA gene (bacteria and archaea): GTDB SSU, r232 here
Primer trimming
cutadapt, linked adapter; 27F and 1492R by default, both primers required
ASV classifiers
ASV inference
DADA2 denoise-ccs (QIIME 2)
1,000 to 1,600 bases
at most two expected errors
chimera removal
ASV filter: 5 reads in 1 sample by default
BLCA
Bayesian LCA on BLAST hits, per rank bootstrap
16S rRNA gene default
Naive Bayes
genus by assignTaxonomy, species by exact match
read level profiler
Length filter
1,000 to 1,600 bases; untrimmed reads
Emu
read level profiler
(expectation maximisation)
Fungal ITS: UNITE general release for Fungi, 19.02.2025 here
ITS extraction
ITSxRust extracts the full ITS region; no primer trimming
ASV classifiers
ASV inference
DADA2 denoise-ccs (QIIME 2)
200 to 1,300 bases
at most two expected errors
chimera removal
ASV filter: 5 reads in 1 sample by default
BLCA
Bayesian LCA on BLAST hits, per rank bootstrap
Naive Bayes
single step assignTaxonomy over seven ranks
read level profiler
Length filter
200 to 1,300 bases; extracted ITS reads
EMITS
read level profiler
(expectation maximisation)
ITS default
GTDB: SSU sequences of the current release
UNITE: the supplied release file
Outputs
taxonomy table per classifier
read tracking table
ASV by sample feature table
execution report and parameters.txt